

During the duration of this capstone project, I have learned many things that I wish I knew at the start of this semester. This not only includes information related to Cybersecurity Maturity Model Certification (CMMC) and compliance in general, but also things about the design process and actual project implementations. Some things I am proud to have done while others may have caused more hardships during the process of creating CMMC Fledge. This paper will dive into those pros and cons by exploring the six (6) core processes of an Information Systems project, inception, planning, approval, design, implementation, and deployment. Though these phases may not have been linear, all of these contributed to the output of the project in some way.

The inception phase sets the entire tone for the project. Here I put together my experience from my internship and my desire to build a product that a customer could use and landed on a compliance assistance program. At this point it was still unclear as to what the system would do. After some deliberation with professors, I decided that CMMC was a hot topic within the cybersecurity industry and was something I should look further into. After learning of its upcoming performance I began to use my knowledge I learned from my internship to see what kind of problems I had seen clients face. I realized that getting your foot in the door required either money or a large amount of time. At this point I had settled into the CMMC system that would help small organizations get their foot into the proverbial compliance door. This is where CMMC Fledge was born. Eventually the main idea for this system became a questionnaire to provide accurate guidance. Looking back I believe I understood the problem to a degree to which it is satisfactory. I do not believe I changed anything regarding the initial inception of the project idea.

Looking back, now I realize that planning may have been one of my weakest phases. During this stage I had initially believed that I had a good scope and a defined process and method for the entirety of the project. The plan I had set forth utilized a fairly linear approach, finishing entire phases of the project at a time. Though this idea initially sounded good on paper, in order to meet deliverables and continue

motivation for the project, I eventually developed an Agile-like approach. Additionally, during this phase I was having issues with the scope of my project, even after the proposal was submitted I had to reduce the scope, as I had envisioned a system larger than time would allow. This narrowed the scope from all three (3) CMMC levels to just two (2). This scope change had two main considerations; the first was time and the second was the ability of the system to accurately represent the additional scrutiny for CMMC 13. The issues faced within this plan created additional work and time that could have been repurposed for other use cases. With what I know now, I would create a plan that allows for continuous integration and development, and move away from a phased approach. Finishing a little bit at time with viewable progress was a huge boost to my motivation and effort.

Initial and subsequent project scope and delivery approvals did not pose as big of a challenge as I would have thought. This was due to my laid out plan. The plan proposed was very detailed, but as discussed within the previous section, issues with the plan seem clearer now. Even with the issues, I still believe the original plan was good, but may have not worked best for myself and my work process. Due to this, the plan was approved and put into motion, to begin work.

Careful consideration went into the design of my project. From the start, I kept ease of use and the potential need for future implementations in mind. This was everything from creating readable file structures to providing resources for future improvements. This phase is where I created the design for the database and designed a local environment for implementation. Later down the line the production environment was created, though its initial design was thought of here in order to provide a smooth transition from testing into production. To support these designs, diagrams and user storyboards were created to show off what the system should accomplish. To facilitate this design a GitHub page was created, for consistent workspaces across my various devices. Design changes to the look and feel of the website via user feedback from storyboards were one of the few design changes made. The environments and database fulfilled their needs. The only thing I wish I changed, looking back on it now, would have been to expand the database to query questions. As will be discussed in the implementation phase the plan

and implementation did not involve queried questions, instead questions and recommendations were done straight within the HTML itself. Though I do not believe this added extra work, or greatly added to the maintenance of the system, I believe it may have been a better practice to have done so after discussion with outside stakeholders.

As I moved forward within the project, these last two processes of an Information System Project became more and more interconnected. As for the implementation of the project, it was originally broken into four (4) phases including research, writing, implementation, and testing. Though these phases had distinct features they became daunting on their own. As stated previously, the project slowly adapted an Agile methodology to system implementation. This created a back and forth between the phases. For example, instead of doing the writing for one-hundred and twenty-five (125) controls in one chunk, I did one control family at a time. This allowed me to finish the implementation for one family and test it to have a fully finished piece of the website, without completely finishing the entire application. With this I was able to make small improvements from each implementation to the next, slowly improving the overall feel of the website. I believe that this change from a strictly linear implementation to a more Agile-like approach, greatly improved the implementation of the project, and this step is one that I am proud of taking.

As mentioned previously the deployment process was intertwined with the implementation phase. Once each section, or control family was completed, I was able to push to the production website, to have a working product for outside stakeholders to give feedback on. The feedback was usually taken into consideration and put that section back into the implementation phase. It should be noted that this production environment was hosted within AWS, and some parts of this deployment phase were not within the local implementation phase. This includes but is not limited to AWS Instance deployment, firewall traffic management, SSL/TLS certification set-up, and domain deployment. After the first few deployment phases, I encountered little to no issues, due to the lessons learned there. One clear example of this, which is obvious now, is that Linux file capitalization matters, where Windows does not. Some

files once deployed cause errors because of this, and the initial fix was time consuming. Though once I realized what the issue was and fixed it, I was able to prevent making that mistake again.

As this project comes to its end, I am happy that I have accomplished my goal, I know that I will miss working on it. This has been a majority of my time for the past three months, and I do feel proud of what I have accomplished. I also believe I had learned a lot about the planning and building of a multi-month project, something that I had not done before going into this. As expressed within this paper, I do feel that there are areas of improvements, or things I wish I had done differently. This insight through trial and error, will be extremely valuable. As I move into the Master's program here at CNU, I have already started making plans for the future implementation of this project, as I would like to polish this and deliver it as a product to small businesses or junior consultants to help them achieve their goals within compliance. Though I know I will have lessons learned from every project I accomplish from here until the end of my career, I hope I can take the ones outlined here and mitigate them in future projects, or potentially the future iteration of this project.